

## MATHEMATICS SUBJECTIVE TEST

### Class X (Session 2026 – 27)

#### ANSWER KEY

#### SECTION-A

1. Multiply both sides by  $x$  :  $x^2 + 1 = 3x \Rightarrow x^2 - 3x + 1 = 0$ .  
So  $a = 1$ ,  $b = -3$ ,  $c = 1$ .  
 $a - b + c = 1 - (-3) + 1 = 5$
2. (D) Assertion (A) is false, but Reason (R) is true.  
Here  $\frac{a_1}{a_2} = \frac{4}{2} = 2$ ,  $\frac{b_1}{b_2} = \frac{p}{2}$ ,  $\frac{c_1}{c_2} = \frac{8}{2} = 4$ .  
For  $p = 4$ :  $\frac{4}{2} = \frac{4}{2} \neq \frac{8}{2}$ , which gives an inconsistent system (no solution). So **Assertion A is false. Reason R is a correct standard result, hence true.**
3.  $p(x) = k$  is a constant polynomial. Since the graph coincides with the  $x$ -axis ( $k = 0$ ), every real number is a zero. Hence **infinitely many zeroes.**
4. We have,  
 $(-1)^n + (-1)^8 = 0 \Rightarrow (-1)^n + 1 = 0 \Rightarrow (-1)^n = (-1)$   
On comparing the powers, we get  $n = 1, 3, 5, \dots$   
 $\therefore$  Hence,  **$n$  is any odd number.**
5. We have,  $x^2 - 2ax + (a^2 - b^2) = 0$   
 $\Rightarrow x^2 - ((a + b) + (a - b))x + (a^2 - b^2) = 0$   
 $\Rightarrow x^2 - (a + b)x - (a - b)x + (a + b)(a - b) = 0 \dots [\because a^2 - b^2 = (a + b)(a - b)]$   
 $\Rightarrow x(x - (a + b)) - (a - b)(x - (a + b)) = 0$   
 $\Rightarrow (x - (a + b))(x - (a - b)) = 0$   
 $\Rightarrow \mathbf{x = a + b, a - b}$

#### SECTION - B

6. Pool dimensions:  $(7-2x)(12-2x) = 36$   $4x^2 - 38x + 84 = 36 \Rightarrow 4x^2 - 38x + 48 = 0 \Rightarrow 2x^2 - 19x + 24 = 0$   $(2x - 3)(x - 8) = 0 \Rightarrow x = 3/2 = \mathbf{1.5\ m}$  (rejecting  $x = 8$ )
7. Given, quadratic equation is  $ky^2 - 11y + (k-23) = 0$   
Let the roots of the above quadratic equation be  $\alpha$  and  $\beta$ .  
Now, Sum of roots,  $\alpha + \beta = -(-11)/k = 11/k \dots$  (i)  
and Product of roots,  $\alpha\beta = (k-23)/k \dots$  (ii)

According to the question,

$$\alpha + \beta = \alpha\beta + 13/21$$

$$\therefore 11k = k - 23k + 1321 \dots [\text{From equations (i) and (ii)}]$$

$$\Rightarrow 11k = k - 23k + 1321$$

$$\Rightarrow 11 - k + 23k = 1321$$

$$\Rightarrow 21(34 - k) = 13k$$

$$\Rightarrow 714 - 21k = 13k$$

$$\Rightarrow 714 = 13k + 21k$$

$$\Rightarrow 34k = 714$$

$$\Rightarrow k = 714/34$$

$$\Rightarrow k = 21$$

8. From  $px^2 + qx + r$ :

$$\alpha + \beta = \frac{-q}{p}, \alpha\beta = \frac{r}{p}$$

$$\begin{aligned}\alpha^3\beta + \beta^3\alpha &= \alpha\beta(\alpha^2 + \beta^2) = \alpha\beta[(\alpha + \beta)^2 - 2\alpha\beta] \\ &= \frac{r}{p} \left[ \left( \frac{-q}{p} \right)^2 - 2 \left( \frac{r}{p} \right) \right] = \frac{r}{p^3} (q^2 - 2pr)\end{aligned}$$

9. Raunak is correct.

Let  $\sqrt{2}$  be a rational number such that  $\sqrt{2} = \frac{a}{b}$ , where  $a$  and  $b$  are coprime and  $b \neq 0$

$$\sqrt{2}b = a$$

$$2b^2 = a^2$$

2 divides  $a^2$ , so 2 divides  $a$  as well.

Let  $a = 2p$  (for some integer  $p$ )

$$a^2 = 4p^2$$

$$2b^2 = 4p^2$$

$$b^2 = 2p^2$$

2 divides  $b^2$ , so 2 divides  $b$  as well.

$\therefore$  2 is a common factor of  $a$  and  $b$  which is a contradiction as  $a$  and  $b$  are coprime.

$\therefore$  Our assumption is wrong. Hence,  **$\sqrt{2}$  is an irrational number.**

10. The number of rooms will be minimum if each room accommodates a maximum number of teachers. Since in each room the same number of teachers are to be seated and all of them must be of the same subject. Therefore, the number of teachers in each room must be HCF of 48, 80, and 144. The prime factorisations of 48, 80 and 144 are as under

$$48 = 2^4 \times 3^1$$

$$80 = 2^4 \times 5^1$$

$$144 = 2^4 \times 3^2$$

$$\therefore \text{HCF of } 48, 80 \text{ and } 144 = 16$$

Therefore, in each room 16 teachers can be seated.

$$\begin{aligned}
 \therefore \text{No. of room required} &= \frac{\text{Total no. of teachers}}{16} \\
 &= \frac{48+80+144}{16} \\
 &= \frac{272}{16} \\
 &= 17
 \end{aligned}$$

### SECTION - C

11. (i) Total dimensions:  $(12 + 2x) \times (10 + 2x) = 360 \Rightarrow 4x^2 + 44x + 120 = 360 \Rightarrow x^2 + 11x - 60 = 0$   
 (ii) (a)  $(x + 15)(x - 4) = 0 \Rightarrow x = 4 \text{ m}$  (rejecting  $x = -15$ )

**OR**

- (ii) (b) Area of walkway =  $12000 \div 50 = 240 \text{ sq. m}$   
 (iii) Perimeter of lawn =  $2(12 + 10) = 44 \text{ m}$ .
12. (i)  $(18 + n)(12 + n) = 2 \times 18 \times 12 = 432$   
 (ii)  $n^2 + 30n - 216 = 0$   
 (iii)  $(n + 36)(n - 6) = 0 \Rightarrow n = 6$ . New dimensions :  $24 \text{ cm} \times 18 \text{ cm}$ .

**OR**

$$(18+x)(12+x) = 220 \Rightarrow x^2 + 30x - 4 = 0. D = 916 \text{ (irrational)} \Rightarrow \text{No rational value of } x.$$

13. Let bicycle =  $a \text{ cal/min}$ , walker =  $b \text{ cal/min}$ .
- (i)  $15a + 30b = 435 \Rightarrow a + 2b = 29 \dots(i)$   
 $30a + 40b = 690 \Rightarrow 3a + 4b = 69 \dots(ii)$
- (ii)  $a_1/a_2 = 1/3 \neq b_1/b_2 = 2/4 = 1/2 \Rightarrow \text{Unique solution } \checkmark$
- (iii) (a) Elimination:  $2 \times (i): 2a + 4b = 58$ . Subtract from (ii):  $a = 11 \text{ cal/min}$ .  $b = 9 \text{ cal/min}$ .  
 (iii) (b) Substitution:  $a = 29 - 2b$  into (ii):  $3(29 - 2b) + 4b = 69 \Rightarrow b = 9, a = 11$ .
14. Let the amount invested in scheme A be ₹ $x$  and in scheme B be ₹ $y$  respectively.

According to question,

$$\begin{aligned}
 \frac{x \times 8 \times 1}{100} + \frac{y \times 9 \times 1}{100} &= 1860 \\
 \Rightarrow 8x + 9y &= 186000 \dots (i)
 \end{aligned}$$

$$\begin{aligned}
 \text{and } \frac{x \times 9 \times 1}{100} + \frac{y \times 8 \times 1}{100} &= 1880 \\
 \Rightarrow 9x + 8y &= 188000 \dots (ii)
 \end{aligned}$$

On solving equations (i) and (ii), we get

$$17y = 170000 \Rightarrow y = 10000$$

Substituting the value of  $y$  in equation (i), we get

$$\begin{aligned}
 8x + 9(10000) &= 186000 \\
 \Rightarrow 8x &= 186000 - 90000 \\
 \Rightarrow 8x &= 96000 \\
 \Rightarrow x &= 96000/8
 \end{aligned}$$

$$\Rightarrow x = 12000$$

$\therefore$  Vijay invested ₹12000 in scheme A and ₹10000 in scheme B.

15. (i) Co-ordinates of point A:  
 Point A is on the y-axis, so  $x = 0$ .  
 $p(0) = -0.0025(0) - 0.025(0) + 136 = 136$   
**Point A = (0, 136)**
- (ii) Span of the arch:  
 Span = distance between the two zeroes of  $p(x)$ .  
 Zeroes (from diagram) = 228.5 and  $-238.5$   
 $\text{Span} = 228.5 - (-238.5) = 228.5 + 238.5$   
**Span = 467 m**
- (iii)(a) Sum of zeroes — verify:  
 Zeroes: 228.5 and  $-238.5$   
 $\text{Sum} = 228.5 + (-238.5) = -10$   
 From polynomial:  $-b/a = -(-0.025)/(-0.0025) = -10 \checkmark$   
**Relationship verified.**
- (iii)(b) Values at  $x = 100$  and  $x = -100$ :  
 $p(100) = -0.0025(10000) - 0.025(100) + 136 = -25 - 2.5 + 136 = 108.5$   
 $p(-100) = -0.0025(10000) - 0.025(-100) + 136 = -25 + 2.5 + 136 = 113.5$   
 $108.5 \neq 113.5$   
 **$\therefore$  Answer:  $p(100) = 108.5$  and  $p(-100) = 113.5$ . They are NOT the same.**

— End of Answer Key —

